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ADMNo. 147EE000907

B. Tech. Common, Semester I

SESSION: 2014-2015

INDIAN SCHOOL OF MINES, DHANBAD

MID SEMESTER EXAMINATION (MONSOON SEMESTER)

Subject: Electronics Engineering

Time: 2 Hours

Max Marks: 60

1. Do not write anything on the question paper.
2. Make suitable assumptions, if necessary and state them clearly. Attempt all questions in order.
3. There are 05 questions in total on two printed pages.

- 1(a). Determine the conductivity of intrinsic germanium wafer at 300°K . If arsenic impurity is added to the extent of 1 impurity atom in 10^7 Ge atoms, Find (i) the ratio of majority to minority carrier concentration (ii) conductivity of the doped semiconductor (iii) Is this p- or n-type germanium? ($n_i = 2.5 \times 10^{13} \text{ cm}^{-3}$, $\mu_p = 1800 \text{ cm}^2/\text{Vsec}$ and $\mu_n = 3800 \text{ cm}^2/\text{Vsec}$) [6]
- 1(b). Explain, How does the dynamic resistance of a semiconductor diode vary with diode current and temperature? Calculate the dynamic resistance for Ge diode having a forward bias voltage of 200 mV at 77°C . A germanium diode at room temperature has a reverse saturation current of $1 \mu\text{A}$. [6]
- 2(a). What is meant by Diffusion and Transition Capacitance of p-n junction diode? Show that the Transition capacitance is inversely proportional to the depletion-layer width. [6]
- 2(b). Explain How does the reverse saturation current of a p-n junction diode vary with temperature and also explain the effect of temperature on diode voltage (at constant current). [6]
- 3(a). Write down the necessary condition for clamping action. For the given input, Sketch output waveform for circuit shown in Fig. 1. [6]

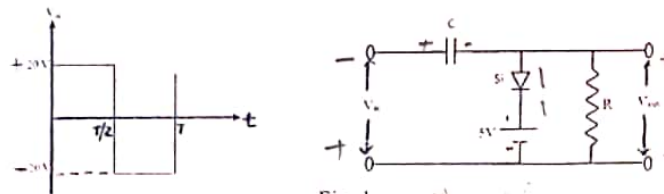


Fig. 1

- 3(b). Explain, Zener diode used as a voltage regulator. Consider the voltage regulator circuit shown in Fig. 2 where Zener diode with breakdown voltages $V_Z = 50 \text{ V}$. Determine the maximum and minimum value of zener diode current I_Z . [6]

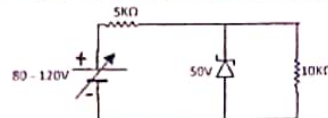


Fig. 2

- 4(a). Two Zener diodes Z_1 and Z_2 connected back-to-back across the input peak to peak voltage of 50V is shown in Fig. 3. Both Zener diodes have zener voltage of 20V. [6]



- (i) Draw the output waveform for the given input waveform.
 (ii) If the peak to peak input voltage is increased to 60V and zener diodes are replaced by other zener diodes, each having zener voltage of 10V. Sketch the output waveform.

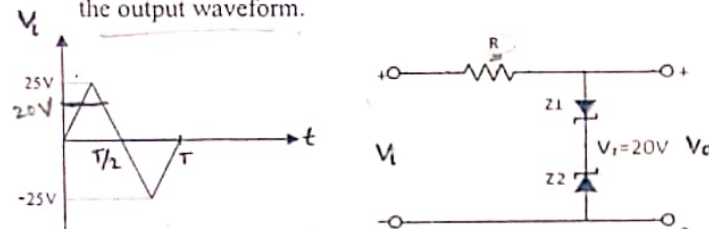
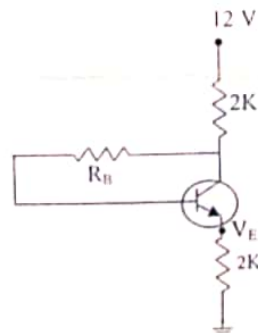


Fig. 3

- 4(b). For a bridge rectifier the load resistance is $1\text{ K}\Omega$, primary to secondary turns ratio of the transformer is 4:1 and the supply to the primary is $240 \sin(140\pi t)\text{ V}$. Each diode has a forward resistance of 100Ω . Calculate (i) Average and effective value of load current (ii) Ripple factor (iii) Efficiency of rectifier. [6]
- 5(a). Explain Early Effect with the aid of plots of potential and minority concentration throughout the base region and also elucidate three consequences of the Early Effect in bipolar junction transistors. [6]
- 5(b). In Fig. 5, the transistor has $\beta = 19$. If $V_E = 4\text{ V}$, Determine the value of R_B . [6]



OR

Explain the various current components in p-n-p transistor with suitable diagram.

SEMESTER: WINTER, SESSION: 2010-2011

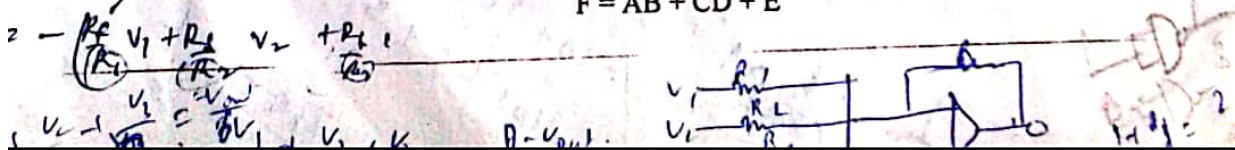
Examination & Semester: B. Tech., I Sem., Common
Subject: ELECTRONICS ENGINEERING

Time: 3 Hours
Max Marks: 100

Instructions: Answer all questions from Part – I and any two from Part – II.

PART – I

- Q.1(a) Discuss the effect of temperature on I-V characteristics (forward and reverse) of a silicon p-n junction diode with suitable graphs. (5)
- (b) A silicon diode has reverse saturation current of $2.5 \mu\text{A}$. Find the forward voltage for a forward current of 10 mA . (4)
- (c) What is the major difference between a bipolar and a unipolar transistor? (1)
- Q.2(a) A full-wave bridge rectifier (using silicon diodes) with $120 \text{ V (V}_{\text{RMS}})$ sinusoidal input has a load resistor of $1 \text{ K}\Omega$. (8)
- What is the dc voltage available at the load?
 - Determine the required PIV rating of each diode
 - Find maximum current through each diode during conduction
 - What is the required power rating of each diode?
- (b) A Zener diode has a breakdown voltage of 5.2 V and maximum power rating of 260 mW . Find the maximum allowable current through the Zener when it is acting as a regulator. (2)
- Q.3(a) In the common emitter circuit shown in Fig.1, an npn transistor having a value of $\beta = 50$ is used with $V_{\text{CC}} = 10 \text{ V}$ and $R_{\text{C}} = 2 \text{ K}\Omega$. If a $100 \text{ K}\Omega$ resistance is connected between collector and base and $V_{\text{BE}} \approx 0$, determine (7)
- I_{CQ} and V_{CEQ}
 - Stability factor $S(I_{\text{CO}})$.
- (b) How do the internal transistor capacitances affect the high frequency response of an amplifier? (3)
- Q.4(a) Explain the mechanism of Zener and avalanche breakdown. (5)
- (b) Given $I_{\text{DSS}} = 8 \text{ mA}$ and $V_{\text{P}} = -4.5 \text{ V}$, sketch the transfer characteristics for the JFET. (5)
- Q.5(a) Sketch the circuit of summing amplifier using op-amp to get (5)
- $$V_{\text{out}} = -(V_1 + 2V_2 + 3V_3)$$
- where V_1 , V_2 , and V_3 are the three inputs to the op-amp.
- (b) Derive the expression for the gain of op-amp based inverting amplifier. (5)
- (c) Design a non-inverting amplifier with a gain of 2. It is given that at the output voltage of 10 V , the current in the voltage divider is to be $10 \mu\text{A}$. (5)
- Q.6 Implement the following function using NAND gates only: (5)
- $$F = AB + CD + E$$



Continued from page - 1

PART - II

Q.7(a) Illustrate the CB and CE modes of operation by drawing the circuit diagrams for a p-n-p and n-p-n transistor. Clearly indicate the directions of current flow. Sketch the CE output characteristics of the transistor and explain the active, saturation and cutoff regions. (10)

(b) A transistor with $\beta=49$ is connected in voltage divider configuration as shown in Fig.2. Given $V_{CC} = 10V$, $R_C = 1 k\Omega$, $V_{CE} = 5 V$, $I_C = 4.9 mA$, and $V_{BE} = 0.2 V$. The stability factor $S(I_{CO})$ is desired to be 10. Obtain the values of R_1 , R_2 and R_E . (10)

Q.8(a) A junction transistor has the following h-parameters: $h_{ie} = 2 K\Omega$, $h_{re} = 1.6 \times 10^{-4}$, $h_{fe} = 50$, $h_{oe} = 50 \mu A/V$. Determine the current gain, voltage gain, input resistance and output resistance of CE amplifier if the load resistance is $12 K\Omega$. (10)

(b) Differentiate between the working principles of an enhancement type and a depletion type MOSFET with suitable I-V characteristics. (10)

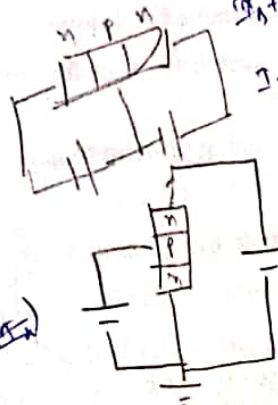
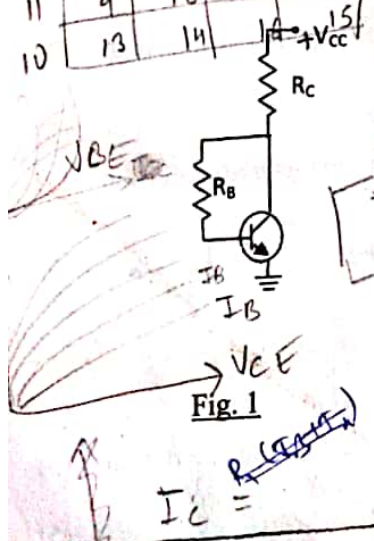
Q.9(a) Simplify the given Boolean expression using Boolean algebra and also 4-variable K-map. Express the function in canonical SOP and POS forms. Draw the gate implementation for the simplified expression. (10)

$$F = A\bar{B}C + \bar{B}\bar{C}D + BCD + AC\bar{D} + \bar{A}\bar{B}C + \bar{A}B\bar{C}D$$

(b) Discuss the operation of S-R latch. How can S-R latch be converted to a D latch. How can you make a D flip-flop using two D latches? (10)

Handwritten K-map for 4 variables (A, B, C, D):

CD \ AB	00	01	11	10
00	1	2	11	3
01	5	6	9	7
11	9	10	12	11
10	13	14	15	16



Handwritten calculation: $2^4 = 16$

Fig. 2

Handwritten calculations for Fig. 2:

$$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_1 + R_2 + \beta(R_C + R_E)}$$

$$I_{CQ} = \beta I_{BQ}$$

$$V_{CEQ} = V_{CC} - I_{CQ}(R_C + R_E)$$

Handwritten calculations for Fig. 1:

$$I_{CQ} = I_{EQ}$$

$$V_{CEQ} = V_{CC} - I_{CQ}R_C$$

Time : 2 Hours

Instructions: All the questions are compulsory.
(The right side numbers indicate marks.)

1. (a) Explain the modifications in Energy Band diagrams for forward and reverse biased p-n junction diodes with junction capacitance. [6]
(b) Define cutin voltage of a diode. [2]
(c) Obtain and explain the expressions for different types of resistances in a biased p-n junction diode. [2]
2. (a) Explicate the I-V characteristics of a forward and reverse biased p-n junction diode. [4]
(b) Explain the static and dynamic characteristics of a diode for a given load resistance. [6]
Describe the transfer characteristics of a diode circuit.
3. (a) A symmetrical square wave voltage of Figure 1 is impressed at the input of the clipping circuit of Figure 2. Assume that for the diode, the cutin voltage $V_i = 0$, the forward resistance $R_f = 0$, and the reverse resistance $R_r = 2 \text{ M}\Omega$. Taking $R = 1 \text{ M}\Omega$ and $V_R = 2.5 \text{ V}$, sketch the o/p voltage (V_o) waveform showing the voltage levels. [6]

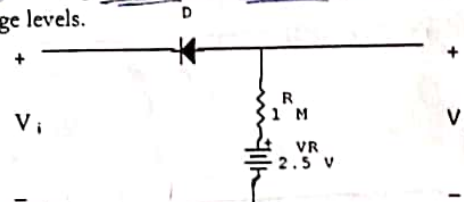
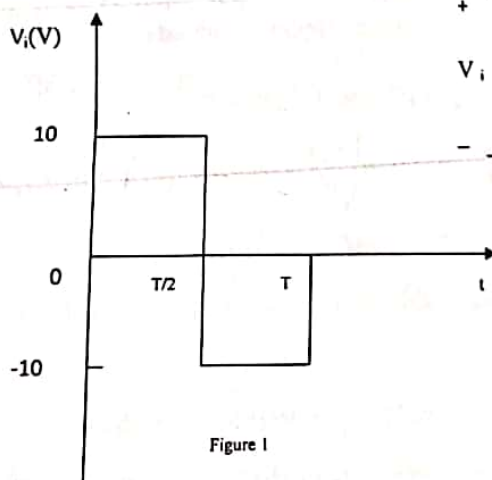
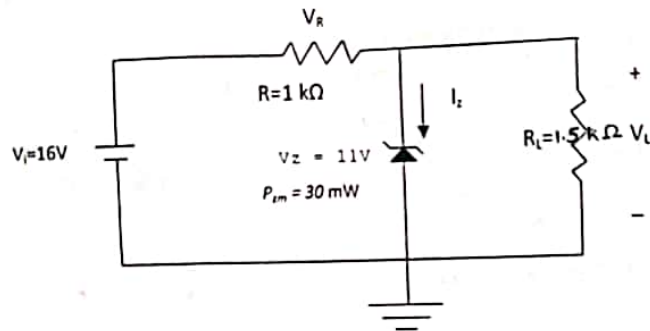


Figure 2

- (b) Explain the operation of Clamping Circuit with diagram. [4]
4. (a) Explain the operation of a full wave rectifier with waveform and obtain its expressions for γ and η . [5]
(b) With circuit diagram explain the working of a bridge rectifier. [5]
5. (a) What are the different breakdowns in diode? [4]
(b) Explain the operation of zener diode and its uses. [6]

(Turn Over)

6. (a) Determine the values of V_L , V_R , I_Z and P_Z for the network containing zener diode as shown below:

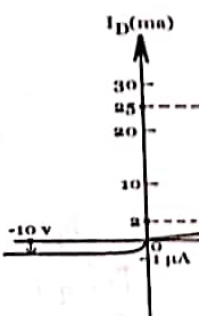
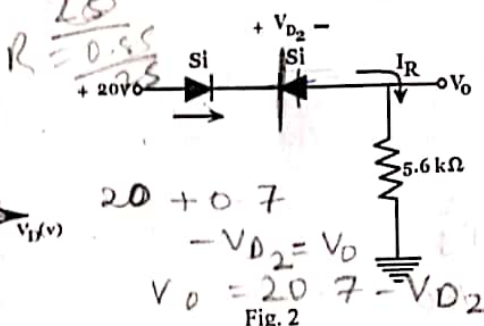
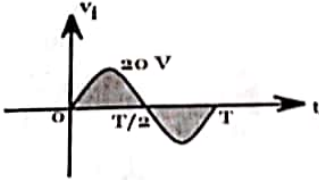
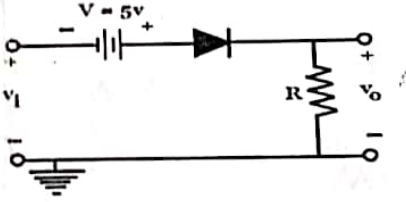
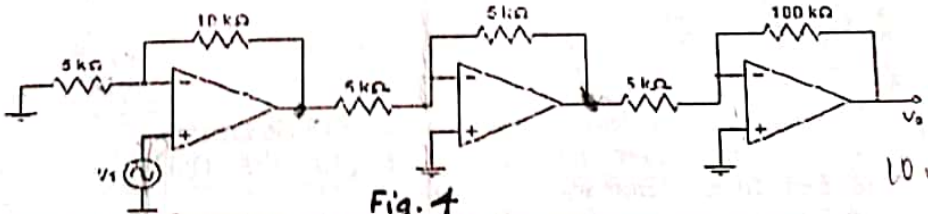


- (b) Repeat part (a) with $R_L = 4\text{ k}\Omega$.

- (c) With Energy band diagram explain the operation of a n-p-n transistor after applying bias voltages.

Instructions: Answer all questions from Part - I and any two from Part - II.

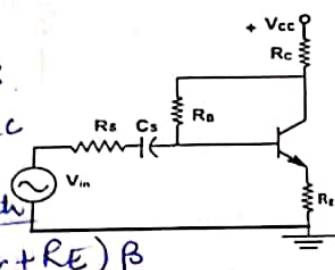
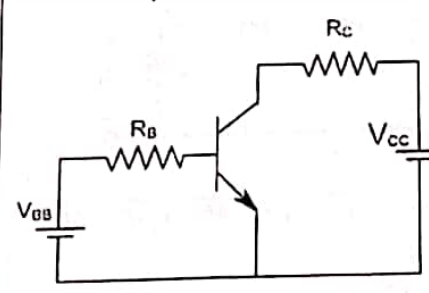
PART - I

Q.1(a)	Determine the dc resistance levels for the diode of Fig.1 at i) $I_D = 2 \text{ mA}$ (low level) ii) $I_D = 25 \text{ mA}$ (high level) iii) $V_D = -10 \text{ V}$ (reverse - biased):  Fig. 1  Fig. 2	(5)
(b)	Determine I_R, V_{D2} and V_O for the circuit of Fig 2.	(5)
Q.2(a)	What is virtual ground and CMRR of an OPAMP? Explain the circuit of Integrator and unity gain buffer using OPAMP.	(5)
(b)	Minimize the following function using K-map & also implement the function using NAND gate: $F(A,B,C,D) = \sum m(4,5,6,10,11,12) + \sum d(2,3,7)$	(5)
Q.3(a)	Determine the output waveform for the sinusoidal input of Fig. 3   Fig. 3	(5)
(b)	Simplify the following Boolean expression: i) $Y(A,B,C) = ABC + \bar{A}\bar{B} + ABC$ ii) $Y(A,B,D) = (\bar{A} + B)(A + B + D)\bar{D}$	(5)
Q.4(a)	In Fig. 4, calculate the input voltage if the final output is 10.08 V.  Fig. 4	(5)

Continued from page - 1

	Fig. 4	(5)
(b)	In Fig. 4 above, calculate the output of the second stage op-amp if $V_1 = 25 \text{ mV}$.	(5)
Q.5(a)	Explain the following terms for an OP-AMP: i) voltage gain, ii) input bias current, iii) input offset current, iv) input offset voltage and v) output offset voltage.	(5)
(b)	The following information is included on the data sheet for an N-channel JFET: $I_{DSS} = 20 \text{ mA}$, $V_P = -8 \text{ V}$ and $g_{m0} = 5000 \mu\text{S}$. Determine the values of drain current and transconductance at $V_{GS} = -4 \text{ V}$.	(5)
Q.6(a)	State few differences between JFET and MOSFET.	(5)
(b)	How do the internal transistor capacitances affect the high frequency response of an amplifier?	(5)

PART - II

Q.7(a)	For the circuit shown in Figure 5 below, $V_C = 5 \text{ Volt}$, $R_C = 10 \text{ K}\Omega$, $R_E = 0.27 \text{ K}\Omega$. Find value of R_B . Consider $\beta = 45$.	(10)
	 Fig. 5	
(b)	Determine the voltage V_0 for the network of Fig. 6	(10)
Q.8(a)	Derive the expression for stability factor in Fig 7, in terms of R_B , R_C & β .	(10)
	 Fig. 7	
(b)	Determine the currents I_1 , I_2 and I_{D2} for the network of Fig. 8	(10)
Q.9	Write short notes on the following topics: (a) Design a Full Adder using two half adder and logic gates (b) Enhancement Type MOSFET versus Depletion Type MOSFET (c) Early Effect or Base-width Modulation (d) 2's complements representation of number	(20) (5 x 4)

Instructions: Answer all questions from Part – I and any two from Part – II.

PART – I

- Q.1(a) Design a clamper circuit using diode to get the output V_o from the input V_i as shown in Figure 1. The diode may be assumed to be ideal. (5)

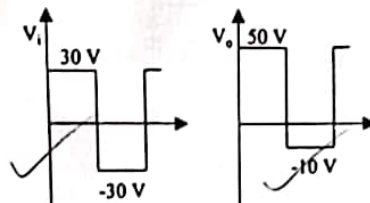
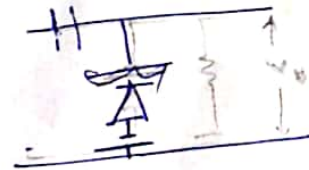


Figure 1



- (b) Determine V_L , I_L , I_Z , and I_R for network shown in Figure 2 if $R_L = 180\Omega$. Determine the value of R_L that will establish maximum power conditions for Zener diode. Determine the minimum value of R_L to ensure that Zener diode is in the "on" state. (5)

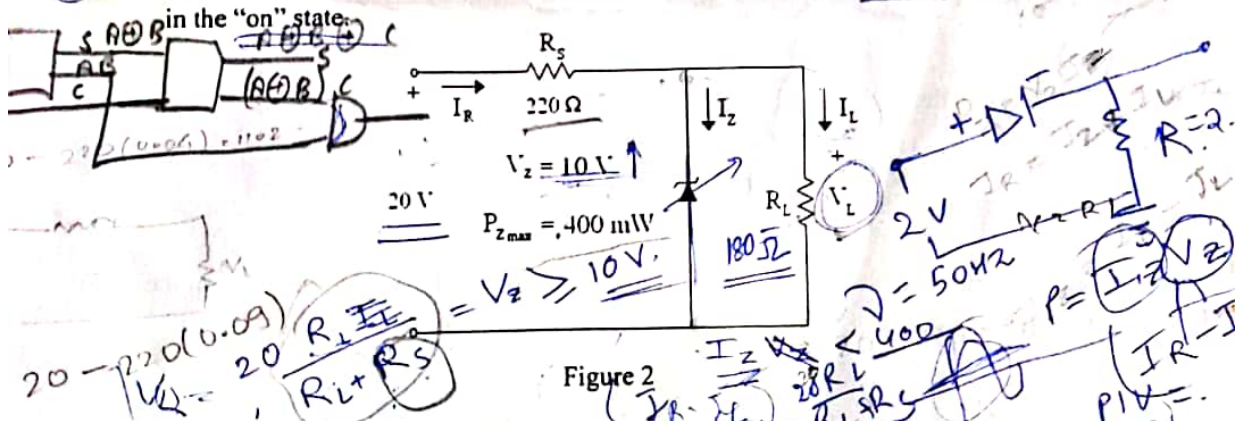


Figure 2

- Q.2(a) A dc voltage of 2V is available across a load of $2.2\text{ K}\Omega$ from a half wave rectifier using an ideal diode. The input is a sinusoidal waveform with a frequency of 50 Hz (5)
- Sketch the input and output waveforms.
 - Determine the required PIV rating of the diode.
 - What is the ripple factor and time period of the output waveform?
 - How will the dc level of the output change if the ideal diode is replaced by a silicon diode?

- (b) For a BJT the common base current gain $\alpha = 0.98$ and the collector base junction reverse bias saturation current $I_{CO} = 0.6\mu\text{A}$. This BJT is connected in the common emitter mode and operated in the active region with a base drive current $I_B = 20\mu\text{A}$. Determine the collector current I_C for this mode of operation. (5)

Continued from page - 1

- Q.3(a) In the circuit shown in Figure 3, $R_S = 1\text{ k}\Omega$, $V_{CC} = 20\text{ V}$, $R_B = 430\text{ k}\Omega$, $R_C = 2\text{ k}\Omega$, $R_E = 1\text{ k}\Omega$. Determine I_B , I_C , and V_{CE} , if the transistor has $\beta = 50$. Verify that the transistor is in active region as required. (4)

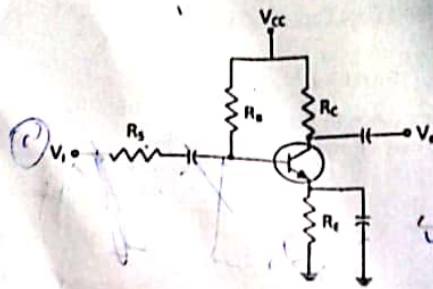


Figure 3

- (b) Define the set of four Hybrid parameters for any two port network stating their units. Sketch the hybrid parameter equivalent model of the amplifier circuit of Figure 3. (6)
- Q.4(a) BJT is a 'bipolar' current-controlled device whereas FET is unipolar and voltage controlled. Explain. (4)
- (b) Define the following (6)
- (i) CMRR in OPAMP
 - (ii) Pinch-off voltage in JFET
 - (iii) Stability factor in BJT amplifiers

- Q.5(a) For the truth table given. Write the logic equation for the output f , in terms of the three inputs A, B , and C , in both SOP as well as POS form. (5)

A	B	C	f
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

SEMESTER: WINTER, SESSION: 2010-2011

Examination & Semester: B. Tech., II Sem., Common
Subject: ELECTRONICS ENGINEERING

Time: 3 Hours
Max Marks: 100

Instructions: Answer all questions from Part - I and any two from Part - II.

PART - I

- Q.1(a) For the circuits shown in Fig. 1 using ideal diodes, find the values of I and V_o . (4)
(b) A diode for which the forward voltage drop is 0.5 V at 1.0 mA is operated at 0.7 V . What is the value of the current? (Assume $\eta=1$ and $V_T = 25\text{ mV}$) (3)
(c) What is the origin of the reverse saturation current in a p-n junction diode? Does the reverse saturation current change with the applied reverse bias voltage and the diode temperature? Explain. (3)

Q.2(a) Explain the term peak inverse voltage in connection with a diode rectifier. Is it different for a half-wave rectifier, a full-wave bridge rectifier and a full-wave rectifier with center-tapped transformer? (3)

(b) Draw a diode clamping circuit. What is the main requirement for its proper clamping action? (3)

(c) Derive the expression of rectification efficiency for a half wave rectifier. (4)

Q.3(a) In the CE configuration of BJT, does the output current depend on the output voltage? Explain. (3)

(b) If $\alpha = 0.98$ for a BJT, find I_B for $I_E = 40\text{ }\mu\text{A}$. (3)

(c) Define I_{CBO} and I_{CEO} . How are they related? (3)

(d) Determine the saturation current ($I_{C_{sat}}$) for the network of Fig. 2. (3)

Q.4(a) Draw and explain the ac equivalent circuit for the network of Fig. 2. (4)

(b) Draw the hybrid equivalent model of a BJT and explain the meaning of the hybrid parameters. (3)

Q.5(a) Calculate the output voltage V_o for the given OPAMP based circuit. (3)



(b) Draw the circuit for an OPAMP based differentiator and derive the equation for output voltage. (3)

(c) Draw the schematic structure for an n-channel JFET and discuss its drain characteristics. (3)

(d) List the truth table of the function $F = xy + xy' + y'z$ and draw its gate implementation. (3)

(e) Prove the following identity: $(A+B)(A+C) = A+BC$ (3)

$$xy + xy' + y'z$$

$$\frac{V_o}{V_i} = - \frac{100 \times 10^3}{22}$$

$$A + AC + BA + BC$$

PART - II

- Q.7(a) Determine I_C and V_{CE} for the network of Fig.2. If the value of β is increased by 50% (i.e., now $\beta = 180$). Determine the percentage change in I_C and V_{CE} . (10)
- (b) Design a fixed bias circuit to have $V_{CE} = 5\text{ V}$ and $I_C = 5\text{ mA}$. The supply voltage is 15 V and $\beta = 100$. If the connections are changed slightly such that the base resistance R_B is connected between the base and the collector of the BJT, what will be the new values of R_B and R_C required to maintain the same levels of I_C and V_{CE} . Can you explain physically why R_B value should be quite different in this case? (10)
- Derive the expressions of current gain (A_i) and voltage gain (A_v) for a transistor amplifier circuit using complete hybrid equivalent model. Also obtain A_i and A_v using approximate hybrid equivalent circuit. (10)
- Describe the principle of n-channel depletion type MOSFET using appropriate diagram. How does it differ from the enhancement type MOSFET in terms of construction and characteristics? (10)
- Write down the truth tables of a half-adder and a full-adder and implement them using logic gates. Support the gate implementations with Boolean expressions. (10)
- Draw the logic diagram for a clocked JK flip-flop and explain its truth table. How can you convert a D flip flop to JK flip-flop? How can you realize a T flip-flop using a JK flip-flop? (10)

$\frac{V_i}{1+h_o R_L}$
 $\frac{-h_f R_L}{1+h_o R_L}$
 $\frac{V_i (1+h_o R_L)}{1+h_o R_L}$
 $\frac{-h_f R_L}{1+h_o R_L}$

$\frac{C_E}{A_i \rightarrow h_{fe}}$
 $A_v \rightarrow -\frac{R_L}{r_{fe}}$

Fig. 1

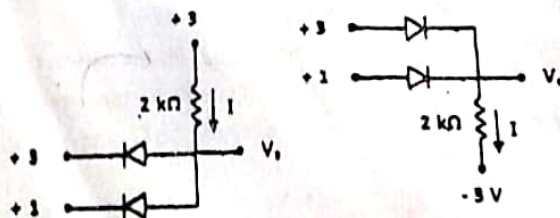
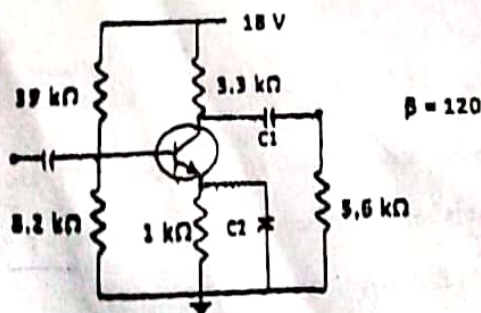


Fig. 2



$Z_o = \frac{h_i - h_f h_o R_L}{1 + h_o R_L}$
 $Z_i \approx h_i$

Examination: B. Tech. Common, Mid Semester

Session: 2013 – 2014

Semester: Monsoon

Time : 2 hours

Subject: Electronics Engineering

Full Marks: 60

(Note: All questions are compulsory)

1. a) At 27°C the intrinsic carrier concentration of Si is $1.5 \times 10^{16} \text{ m}^{-3}$. If the electron and hole mobility are 0.12 and $0.05 \text{ m}^2/\text{Vs}$ respectively, determine the intrinsic resistivity of Si at 27°C . (Given $p_i = 2.315 \times 10^{12} \Omega \text{ m}$) (4)
- b) For the bridge rectifier in the figure below, calculate the following

- (i) Output voltage across the load, (ii) DC output power, (iii) The ripple factor at the output, (iv) The efficiency of rectification and (v) PIV rating for the silicon diodes.

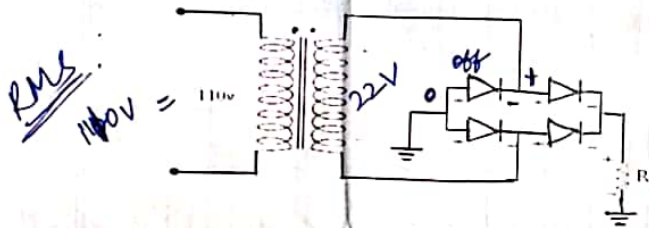


Fig 1

Consider forward bias diode drop of 0.7 V and forward bias resistance of 100Ω for each Si diode. Primary to secondary turns ratio is $5:1$. The value of load resistance is $10 \text{ k}\Omega$. (6)

- c) Prove that the maximum rectification efficiency of a centre-tapped full wave rectifier is approximately double of that of a half-wave rectifier. (2)

2. For the circuit in fig 2, determine the output voltage (V_{out}) and I_c of the circuit when the adjust terminal of potentiometer is at

- (i) Full down position (position C)
- (ii) Exact middle position (position B)
- (iii) Top most position (position A)

Justify your answer for different conditions. ($\beta = 100$)

(12)

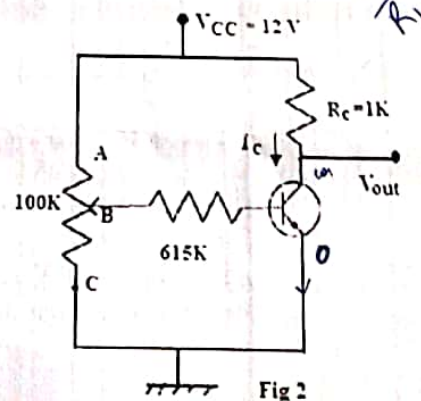
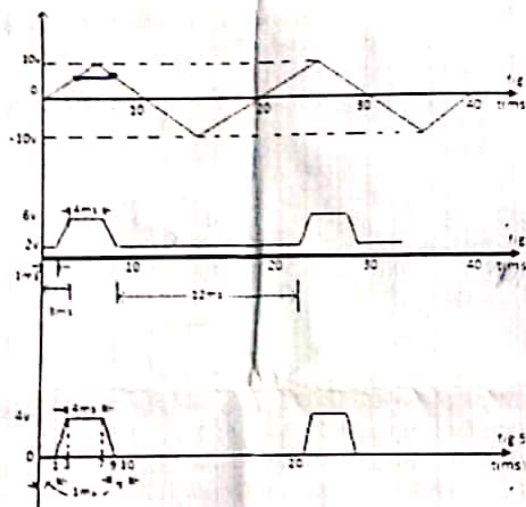


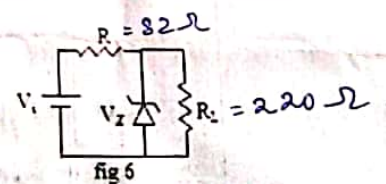
Fig 2

3. a) Explain that (i) at very high temperature extrinsic semiconductor behaves like an intrinsic and (ii) at 0K intrinsic semiconductor behaves like an insulator. (2+2)
- b) Draw the typical I-V characteristics of a diode. Explain the effect of temperature on the forward and reverse characteristics of a pn junction diode. (2+3)
- c) At room temperature, determine the current in a Si diode for forward-bias voltages of 0.6 V if the reverse saturation current is $I_S = 10^{-11}$ A. (3)
4. A voltage source generates an perfectly symmetric triangular waveform having 20ms time period and peak amplitude of 10V as shown in fig. 3 (5)
- a) Design a suitable circuit using ideal diodes having max. current rating of 10 mA and other components to drive the waveform of fig.4 from fig.3.



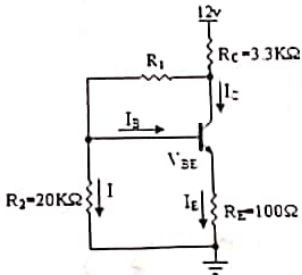
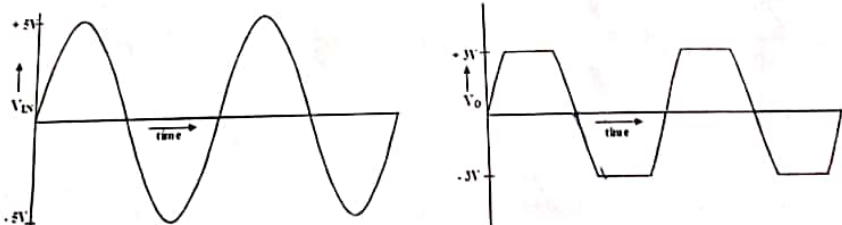
- b) Design suitable circuit using ideal diodes and other components to generate the waveform of fig.5. (5)
- c) Explain the Avalanche and Zener breakdown mechanisms of a p-n junction diode. (2)
5. (a) What is Early effect? Explain its influence in the characteristics of BJT. (3+3)
- (b) How does Zener diode works in voltage regulation circuit? (2+4)

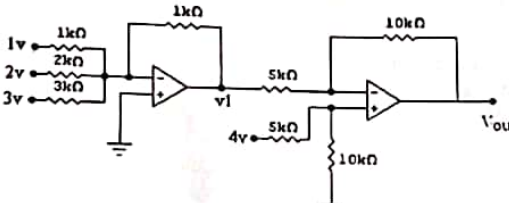
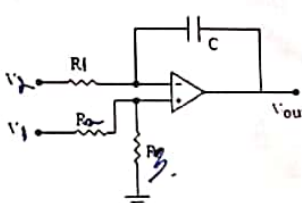
For given the voltage regulator circuit shown in fig 6, with $R = 82\Omega$, $R_L = 220\Omega$, $V_Z = 8\text{ V}$ and $P_{Z_{max}} = 400\text{ mW}$, Determine the allowed range of input voltage (V_i).



$V_Z = 8\text{ V}$

$8 = \frac{220 \times V_i}{220 + 82}$

SEMESTER: MONSOON, SESSION: 2013-2014		
Examination & Semester: B. Tech., I Sem., (Common)		
Time : 3 Hours	Subject: ELECTRONICS ENGINEERING (ECC11101)	Max Marks: 100
Instructions: Answer all the questions from Part I and any two from Part II. (The right side numbers indicate marks.)		
Part I		
1.	(a) Draw and explain the input and output characteristics of a transistor in common emitter mode. Explain how Early effect influences these characteristics. (b) In Fig. 1, $\alpha = 0.98$ and $V_{BE} = 0.7V$. Find the resistance R_1 for an emitter current $I_E = 2mA$. Neglect I_{CO}	4+2 4
 Fig.1		
2.	Design circuits to get the output V_O from the input V_{IN} shown in the Fig.2 using (i) Ideal diode (ii) Si diode.	10
 Fig.2		
3.	(a) What is the effect of temperature on Avalanche and Zener breakdown mechanisms of a p-n junction diode? (b) A Si diode has reverse saturation current, $I_s = 10^{-14} A$ at $25^\circ C$. Find the value of I_s at $125^\circ C$ if I_s increases by 15% per $^\circ C$ rise in temperature. (c) Draw the equivalent circuit of a practical diode and signify forward resistance of a diode.	4 3 3

4.	(a) Calculate the output voltage of the OP AMP circuit as shown in Fig.3 (b) In the circuit of Fig. 4, express V_O in terms of V_1 and V_2.   Fig. 3 Fig. 4	10
5.	A full-wave rectifier uses a double diode, the forward resistance of each element being $100\ \Omega$. The rectifier supplies current to a load resistance of $1\text{ K}\ \Omega$. The primary to total secondary turns ratio of the centre tapped transformer is 10:1. The primary is fed from a supply of 240V(rms). Find the dc load current, the dc output power, the ripple factor, PIV and the efficiency of rectification.	10
6.	(a) Names the gates that are used as Universal gates. Explain how the basic gates can be realized using Universal gates. (b) Compare between FET and BJT	6 4
Part II		
7.	(a) What is thermal runaway? What are the factors that affect the bias stability of a transistor and how these factors change the value of I_C? (b) Derive expression for the stability factors of a voltage divider bias circuit. Hence discuss the choice of the circuit parameters for better stability with respect to I_{CO}, V_{BE}, and β.	10 10
8.	(a) What is pinch-off? Draw a p-channel depletion type MOSFET showing the source, drain, and gate. Explain briefly how does it work and also draw the I_D vs. V_{DS} characteristics. (b) State the characteristics of an ideal OP AMP. What do you mean by virtual ground of an OP AMP? Explain with circuit diagram, the operation of a differentiator using OP AMP.	10 10
9.	(a) What is a half adder? Implement half adder with NAND gates? Give the truth table of a full adder and hence show that a full adder can be constructed using half adders and OR gate. (b) Simplify the following logical expression using Boolean algebra and K map. $F(A, B, C, D) = ABCD + A\bar{B}\bar{C}\bar{D} + A\bar{B}C + AB$	10 10

SEMESTER: MONSOON, SESSION: 2011-2012

Examination & Semester: B. Tech., I Sem., Common
Subject: ELECTRONICS ENGINEERING

Time: 3 Hours
Max Marks: 100

Instructions: Answer all questions from Part – I and any two from Part – II.

PART – I

- Q.1(a) A Si diode has $I_s = 10^{-14}$ A at 25 °C. Find the value of I_s at 125 °C if I_s increases by 15% per °C rise in temperature. (3)
- (b) A Si diode operates at 27 °C. If the applied forward bias is increased, the current is doubled. Calculate the increase in bias voltage. Assume $I_D \gg I_s$ and $\eta = 2$. (4)
- (c) Explain the terms static resistance and dynamic resistance for a semiconductor diode. (3)
- Q.2(a) A full wave bridge rectifier with a 120 V rms sinusoidal input has a load resistor of 1 k Ω . What is the dc voltage available at the load? Determine the required PIV rating of each diode. Find the maximum current through each diode. Assume Si diodes. (6)
- (b) Draw a diode clamping circuit and explain its working. (4)
- Q.3(a) Sketch and explain the input and output characteristics for a BJT in common emitter configuration. (6)
- (b) Consider a transistor with α close to unity. For a small fractional change $\Delta\alpha/\alpha$, show that (4)
- $$\frac{\Delta\beta}{\beta} = \beta \frac{\Delta\alpha}{\alpha}$$
- Q.4(a) Draw a voltage divider bias circuit in common emitter configuration with input and output coupling capacitors and a capacitance parallel to the emitter resistance R_E . Draw and explain the ac equivalent circuit for the same. (6)
- (b) Given a JFET with $I_{DSS} = 6$ mA and $V_P = -4.5$ V (i) Determine I_D at $V_{GS} = -2$ V (4)
(ii) Determine V_{GS} at $I_D = 3$ mA.
- Q.5(a) Calculate the output voltage V_o for the OPAMP based circuit given in Fig.1. (5)
- (b) Draw the circuit for an OPAMP based non-inverting amplifier and derive the expression for voltage gain. (5)
- Q.6(a) List the truth table of the function $F = xy + xy' + y'z$ and draw its gate implementation. (4)
- (b) Draw the logic diagram of RS latch and obtain its truth table. (6)

CORE COURSE STRUCTURE (I & II SEMESTER)				
Name of the Course				
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P				
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	3	1	0	7
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	3	1	0	7
	3	1	0	7
	3	1	0	7
	3	1	0	7

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Page No. - 2

PART - II

- Q.7(a) Consider a voltage divider bias circuit (Fig.2) with the following specifications: $R_1 = 39 \text{ K}\Omega$, $R_2 = 8.2 \text{ K}\Omega$, $R_C = 3.3 \text{ K}\Omega$, $R_E = 1 \text{ K}\Omega$, $V_{CC} = 18 \text{ V}$, $\beta = 120$. Determine I_C , V_{CE} , I_B , V_E , and V_B using the approximate analysis. Repeat the calculations using exact analysis and compare the results. (10)
- (b) Derive the expression of $S(I_{CQ})$ for the voltage divider bias circuit and calculate its value for the circuit used in part (a) of this question. (10)
- Q.8(a) Derive the expressions of current gain (A_i) and voltage gain (A_v) for a CE transistor amplifier circuit using complete hybrid equivalent model. (10)
- (b) Describe the principle and characteristics of an n-channel depletion type MOSFET and enhancement type MOSFET. Use appropriate diagrams. (10)
- Q.9(a) Simplify the given Boolean expression using Boolean algebra and also 4-variable K-map. Express the function in canonical SOP and POS forms. (10)
- $$F(A, B, C, D) = ABC + BCD + A\bar{C}D + \bar{A}B + A.$$
- (b) Write down the truth tables of a half-adder and a full-adder and implement them using logic gates. Support the gate implementations with Boolean expressions. (10)

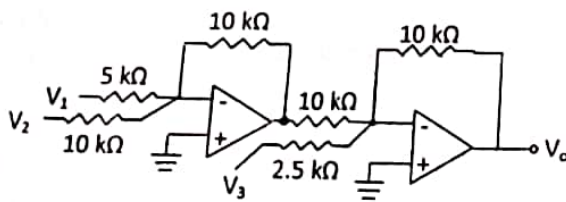


Figure.1

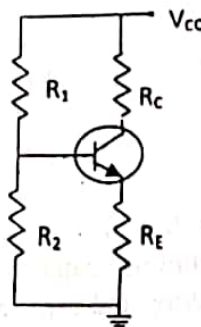


Figure.2